Causal Analysis Report

BPI Challenge 2017 - Offer log.xes.gz-imf

Decision points:	85990
Choice sets:	2
Transitions fitted:	5
Significant (p<0.05):	0
Generated:	2026-06-23 16:04

Step 1 — Association test (logistic regression + permutation)
H□: choice □ util | choice_set.   reject_H0=True → util predicts routing.

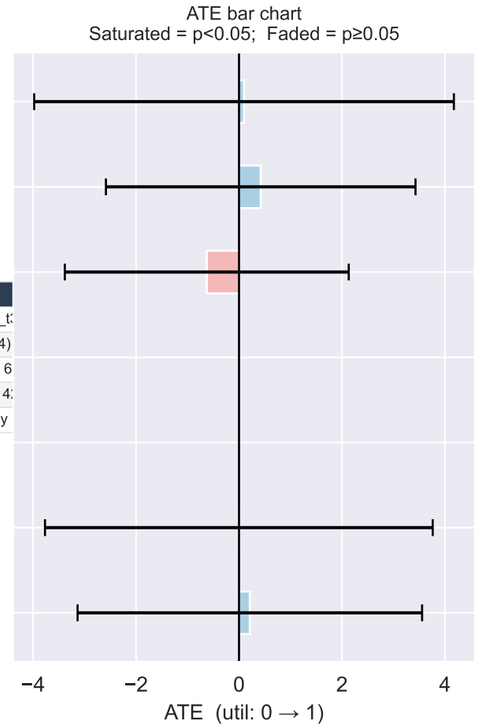
(no data)

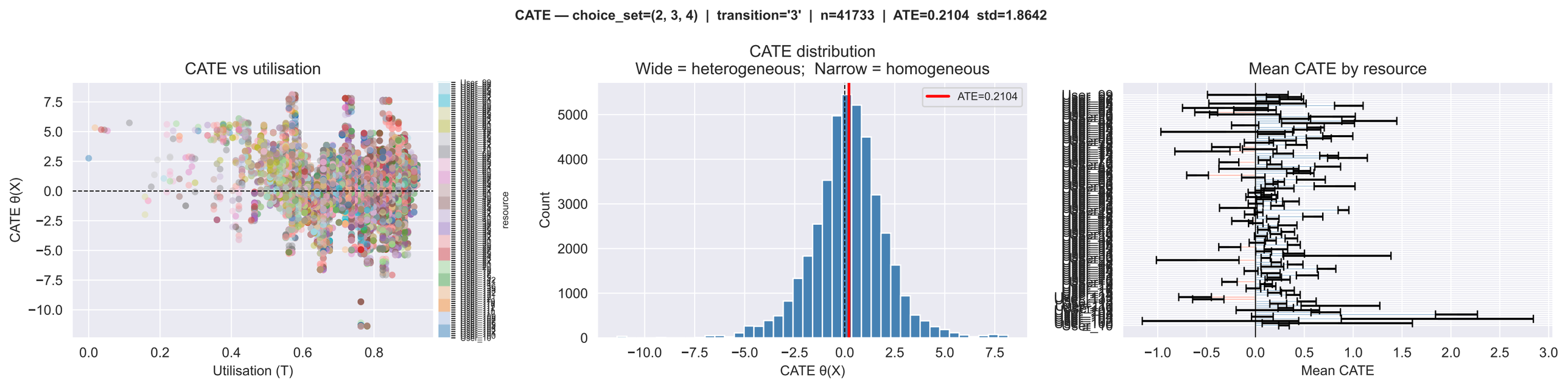
Step 2 — Double ML (partially linear regression, HC3 robust SE)
θ = linear effect of 1-unit ↑ util on P(chose k), after partialling out X.
Equals ATE when the effect is homogeneous. Otherwise: variance-weighted average of CATEs.

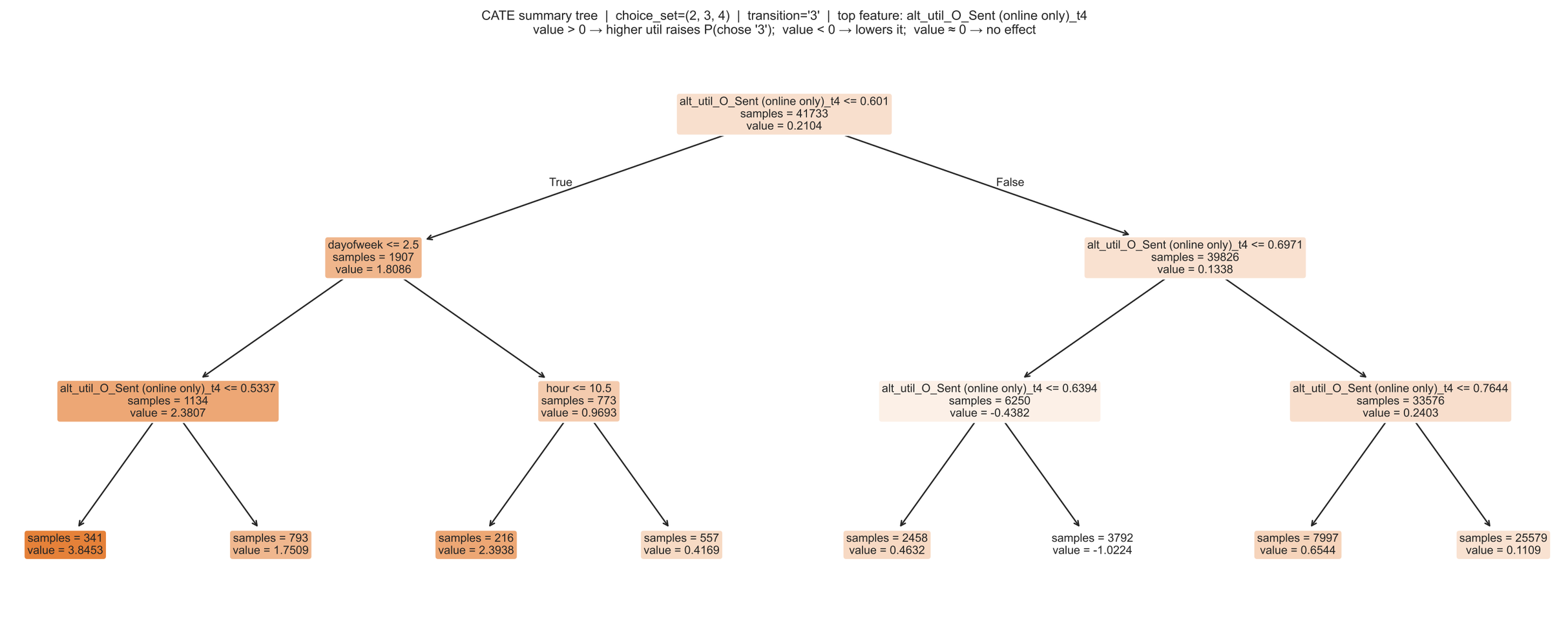
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	theta	se	ci_95	p_value	reject_H0	note
(2, 3, 4)	('τ_t2', 'O_Sent (mail and online)_t3', 'O_Sent (online only)_t4')	41733	3, 4	3	O_Sent (mail and online)_t3	O_Sent (online only)_t4	0.237002	0.082025	[0.0762, 0.3978]	0.0039	True	θ=+0.2370: 1-unit ↑ util ↑ P(t3/O_Sent (mail and online)_t3) by 23.7%
(2, 3, 4)	('τ_t2', 'O_Sent (mail and online)_t3', 'O_Sent (online only)_t4')	41733	3, 4	4	O_Sent (online only)_t4	O_Sent (mail and online)_t3	0.031542	0.091883	[-0.1485, 0.2116]	0.7314	False	θ=+0.0315: 1-unit ↑ util ↑ P(t4/O_Sent (online only)_t4) by 3.2%
(6, 7, 8, 9)	('τ_t6', 'O_Accepted_t7', 'O_Refused_t8', 'O_Cancelled_t9')	42821	7, 8, 9	7	O_Accepted_t7	O_Cancelled_t9, O_Refused_t8	-0.086324	0.034529	[-0.1540, -0.0186]	0.0124	True	θ=-0.0863: 1-unit ↑ util ↓ P(t7/O_Accepted_t7) by 8.6%
(6, 7, 8, 9)	('τ_t6', 'O_Accepted_t7', 'O_Refused_t8', 'O_Cancelled_t9')	42821	7, 8, 9	8	O_Refused_t8	O_Accepted_t7, O_Cancelled_t9	0.366393	0.076225	[0.2170, 0.5158]	0.0	True	θ=+0.3664: 1-unit ↑ util ↑ P(t8/O_Refused_t8) by 36.6%
(6, 7, 8, 9)	('τ_t6', 'O_Accepted_t7', 'O_Refused_t8', 'O_Cancelled_t9')	42821	7, 8, 9	9	O_Cancelled_t9	O_Accepted_t7, O_Refused_t8	0.102819	0.097026	[-0.0874, 0.2930]	0.2893	False	θ=+0.1028: 1-unit ↑ util ↑ P(t9/O_Cancelled_t9) by 10.3%

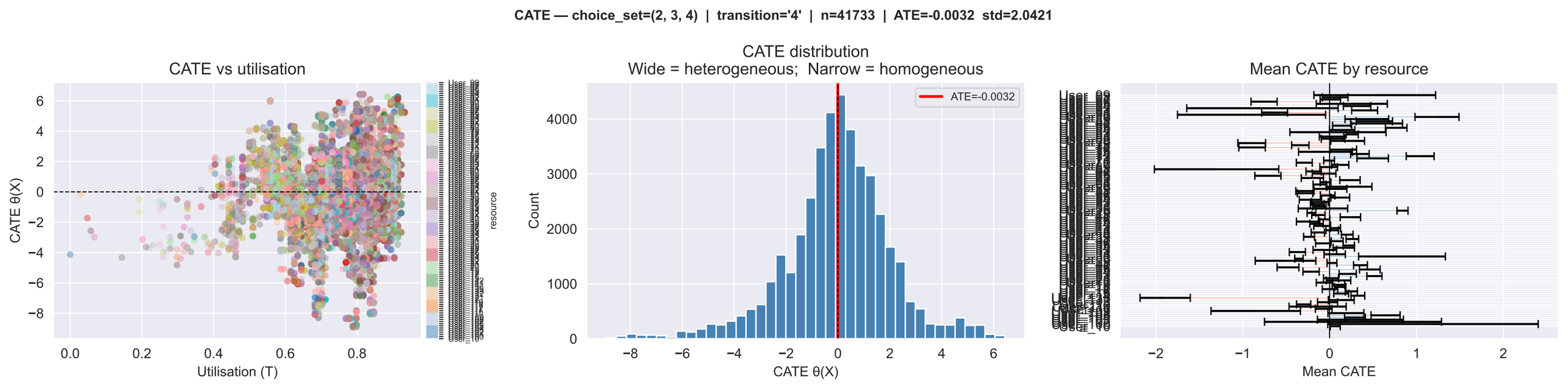
Step 3 — Causal Forest DML (ATE per transition)
ATE = mean of per-sample CATEs. Green rows = H0 rejected (p<0.05).
Less powerful for significance than Double ML, but basis for heterogeneity analysis.

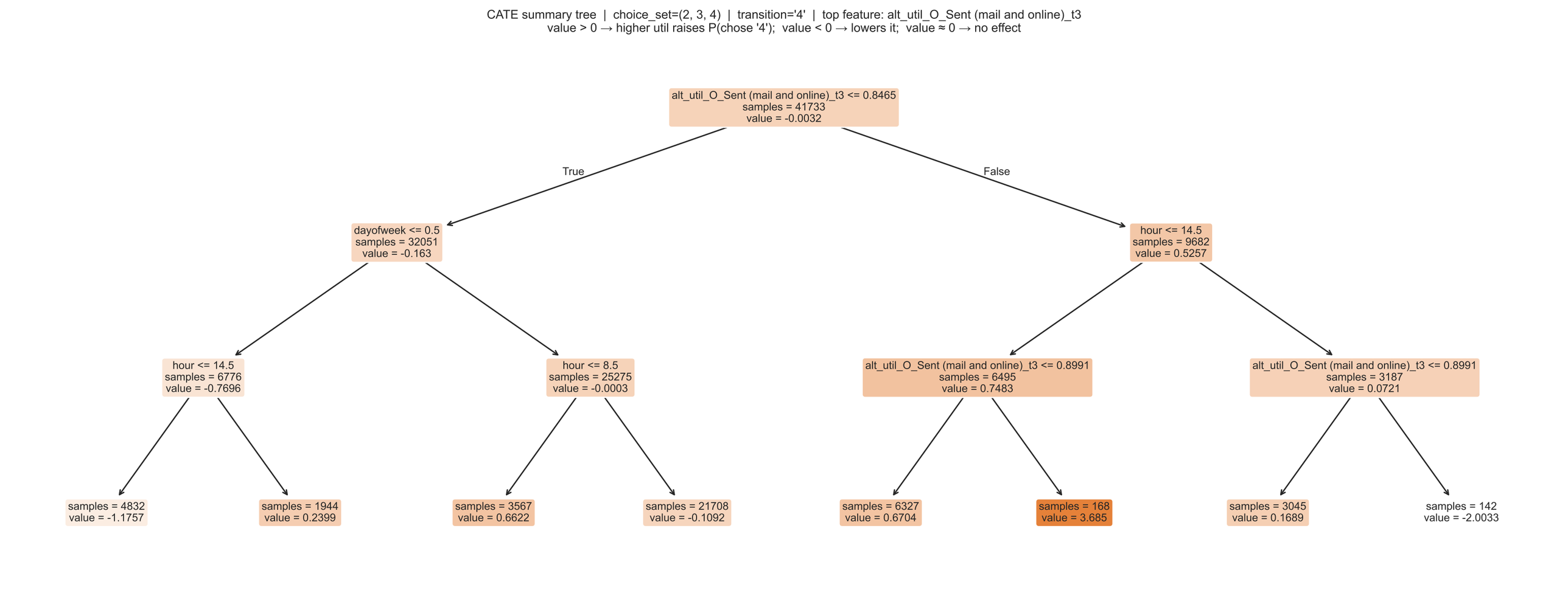
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	ate	ci_95	p_value	reject_H0	note
(2, 3, 4)	('τ_t2', 'O_Sent (mail and online)_t3', 'O_Sent (online only)_t4')	41733	3, 4	3	O_Sent (mail and online)_t3	O_Sent (online only)_t4	0.210368	[-3.1360, 3.5568]	0.9019	False	ATE=+0.2104: 0→1 util ↑ P(t3/O_Sent (mail and online)_t
(2, 3, 4)	('τ_t2', 'O_Sent (mail and online)_t3', 'O_Sent (online only)_t4')	41733	3, 4	4	O_Sent (online only)_t4	O_Sent (mail and online)_t3	-0.003188	[-3.7691, 3.7627]	0.9987	False	ATE=-0.0032: 0→1 util ↓ P(t4/O_Sent (online only)_t4)
(6, 7, 8, 9)	('τ_t6', 'O_Accepted_t7', 'O_Refused_t8', 'O_Cancelled_t9')	42821	7, 8, 9	7	O_Accepted_t7	O_Cancelled_t9, O_Refused_t8	-0.625986	[-3.3836, 2.1316]	0.6564	False	ATE=-0.6260: 0→1 util ↓ P(t7/O_Accepted_t7) by 6
(6, 7, 8, 9)	('τ_t6', 'O_Accepted_t7', 'O_Refused_t8', 'O_Cancelled_t9')	42821	7, 8, 9	8	O_Refused_t8	O_Accepted_t7, O_Cancelled_t9	0.421562	[-2.5851, 3.4282]	0.7835	False	ATE=+0.4216: 0→1 util ↑ P(t8/O_Refused_t8) by 4
(6, 7, 8, 9)	('τ_t6', 'O_Accepted_t7', 'O_Refused_t8', 'O_Cancelled_t9')	42821	7, 8, 9	9	O_Cancelled_t9	O_Accepted_t7, O_Refused_t8	0.097606	[-3.9788, 4.1740]	0.9626	False	ATE=+0.0976: 0→1 util ↑ P(t9/O_Cancelled_t9) by

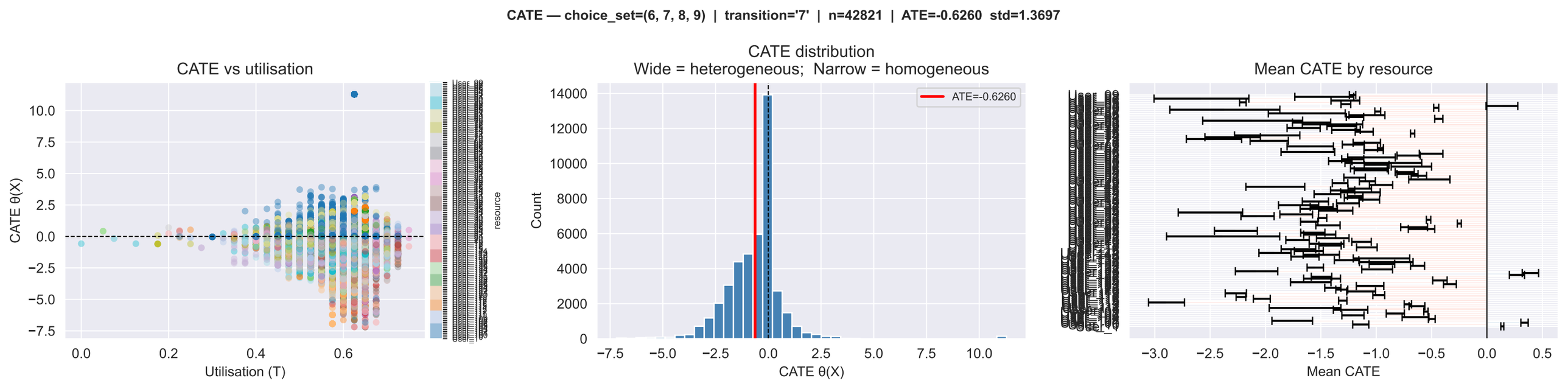


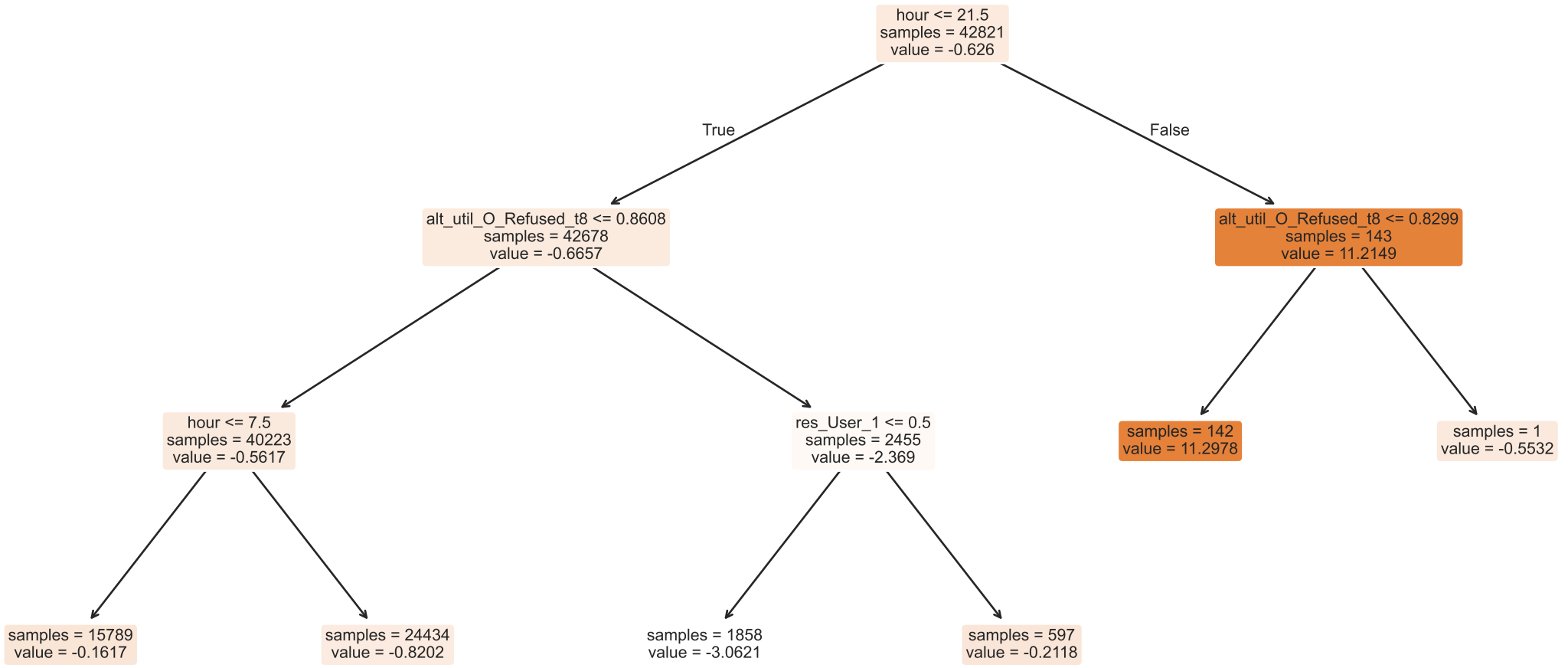

CATE — choice_set=(2, 3, 4) | transition='3' | n=41733 | ATE=0.2104 std=1.8642
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.2104
Count
CATE θ(X)
Mean CATE by resource
User
Mean CATE

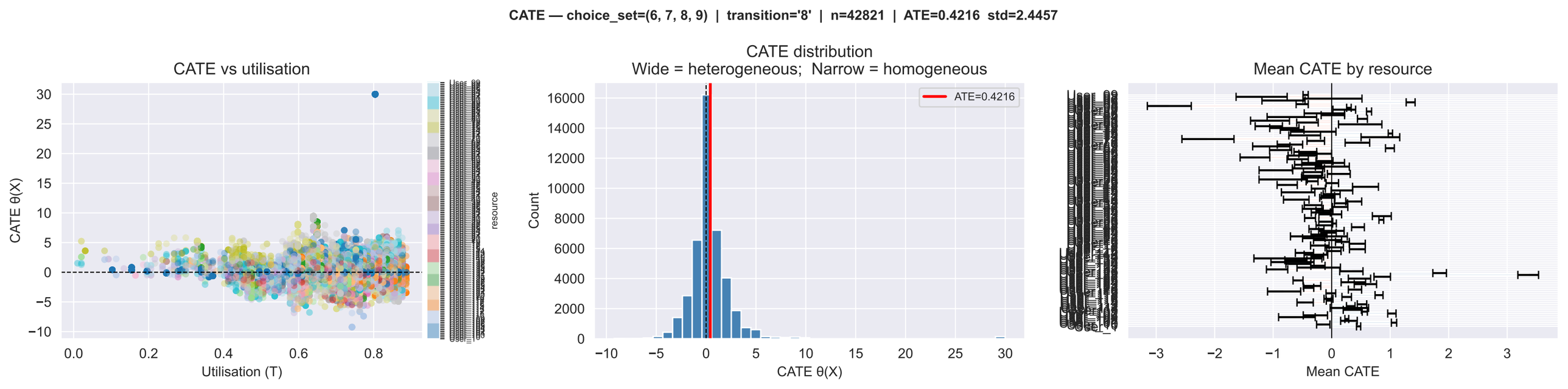

CATE summary tree | choice_set=(2, 3, 4) | transition='3' | top feature: alt_util_O_Sent (online only)_t4
value > 0 → higher util raises P(chose '3'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_O_Sent (online only)_t4 <= 0.601
samples = 41733
value = 0.2104
True
False
dayofweek <= 2.5
samples = 1907
value = 1.8086
alt_util_O_Sent (online only)_t4 <= 0.6971
samples = 39826
value = 0.1338
alt_util_O_Sent (online only)_t4 <= 0.5337
samples = 1134
value = 2.3807
hour <= 10.5
samples = 773
value = 0.9693
alt_util_O_Sent (online only)_t4 <= 0.6394
samples = 6250
value = -0.4382
alt_util_O_Sent (online only)_t4 <= 0.7644
samples = 33576
value = 0.2403
samples = 341
value = 3.8453
samples = 793
value = 1.7509
samples = 216
value = 2.3938
samples = 557
value = 0.4169
samples = 2458
value = 0.4632
samples = 3792
value = -1.0224
samples = 7997
value = 0.6544
samples = 25579
value = 0.1109

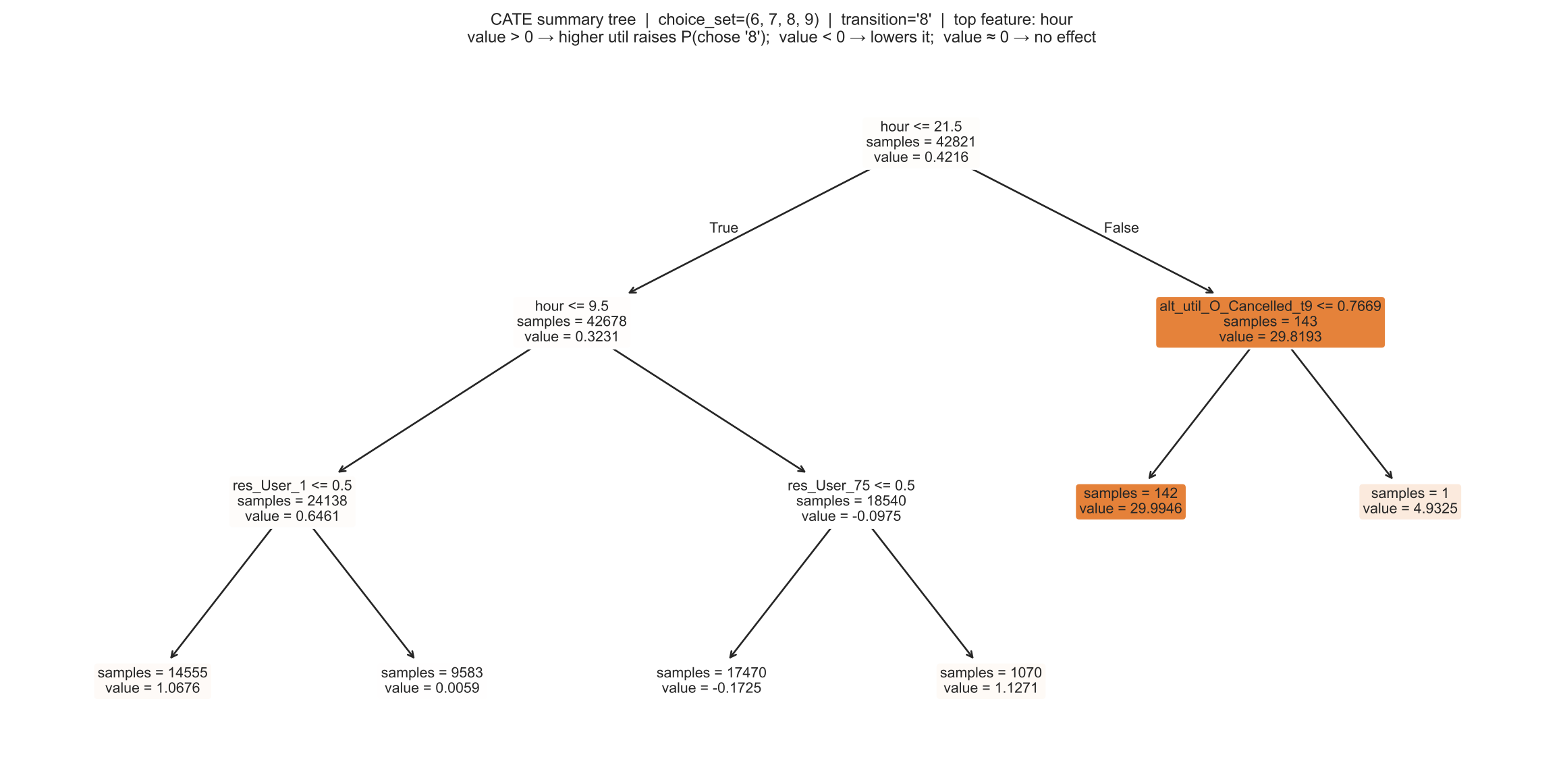

CATE — choice_set=(2, 3, 4) | transition='4' | n=41733 | ATE=-0.0032 std=2.0421
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.0032
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

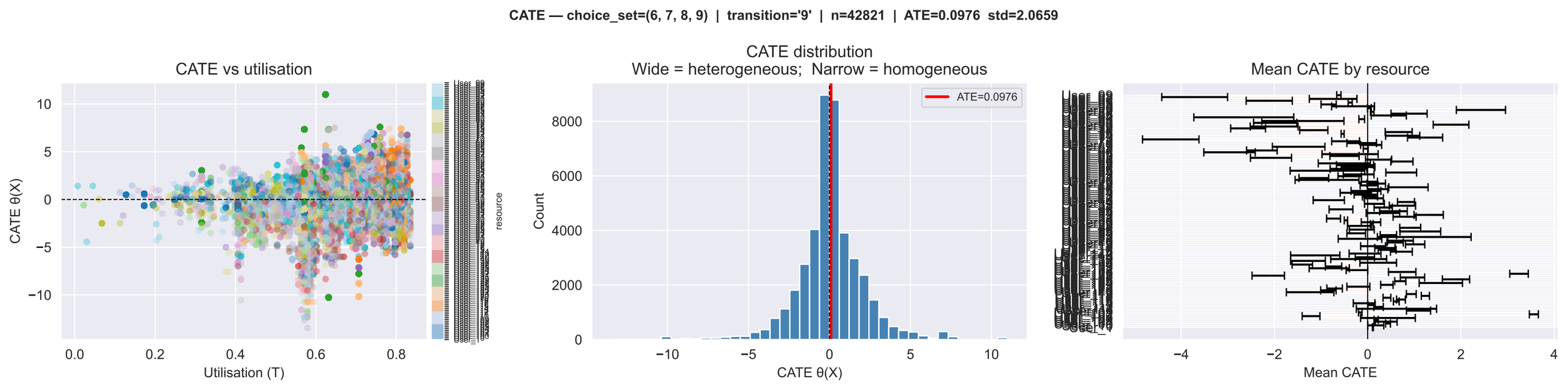

CATE summary tree | choice_set=(2, 3, 4) | transition='4' | top feature: alt_util_O_Sent (mail and online)_t3
value > 0 → higher util raises P(chose '4'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_O_Sent (mail and online)_t3 <= 0.8465
samples = 41733
value = -0.0032
True
False
dayofweek <= 0.5
samples = 32051
value = -0.163
hour <= 14.5
samples = 9682
value = 0.5257
hour <= 14.5
samples = 6776
value = -0.7696
hour <= 8.5
samples = 25275
value = -0.0003
alt_util_O_Sent (mail and online)_t3 <= 0.8991
samples = 6495
value = 0.7483
alt_util_O_Sent (mail and online)_t3 <= 0.8991
samples = 3187
value = 0.0721
samples = 4832
value = -1.1757
samples = 1944
value = 0.2399
samples = 3567
value = 0.6622
samples = 21708
value = -0.1092
samples = 6327
value = 0.6704
samples = 168
value = 3.685
samples = 3045
value = 0.1689
samples = 142
value = -2.0033


CATE — choice_set=(6, 7, 8, 9) | transition='7' | n=42821 | ATE=-0.6260 std=1.3697
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.6260
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(6, 7, 8, 9) | transition='7' | top feature: hour
value > 0 → higher util raises P(chose '7'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 21.5
samples = 42821
value = -0.626
True
False
alt_util_O_Refused_t8 <= 0.8608
samples = 42678
value = -0.6657
alt_util_O_Refused_t8 <= 0.8299
samples = 143
value = 11.2149
hour <= 7.5
samples = 40223
value = -0.5617
res_User_1 <= 0.5
samples = 2455
value = -2.369
samples = 142
value = 11.2978
samples = 1
value = -0.5532
samples = 15789
value = -0.1617
samples = 24434
value = -0.8202
samples = 1858
value = -3.0621
samples = 597
value = -0.2118


CATE — choice_set=(6, 7, 8, 9) | transition='8' | n=42821 | ATE=0.4216 std=2.4457
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.4216
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(6, 7, 8, 9) | transition='8' | top feature: hour
value > 0 → higher util raises P(chose '8'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 21.5
samples = 42821
value = 0.4216
True
False
hour <= 9.5
samples = 42678
value = 0.3231
alt_util_O_Cancelled_t9 <= 0.7669
samples = 143
value = 29.8193
res_User_1 <= 0.5
samples = 24138
value = 0.6461
res_User_75 <= 0.5
samples = 18540
value = -0.0975
samples = 142
value = 29.9946
samples = 1
value = 4.9325
samples = 14555
value = 1.0676
samples = 9583
value = 0.0059
samples = 17470
value = -0.1725
samples = 1070
value = 1.1271


CATE — choice_set=(6, 7, 8, 9) | transition='9' | n=42821 | ATE=0.0976 std=2.0659
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.0976
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

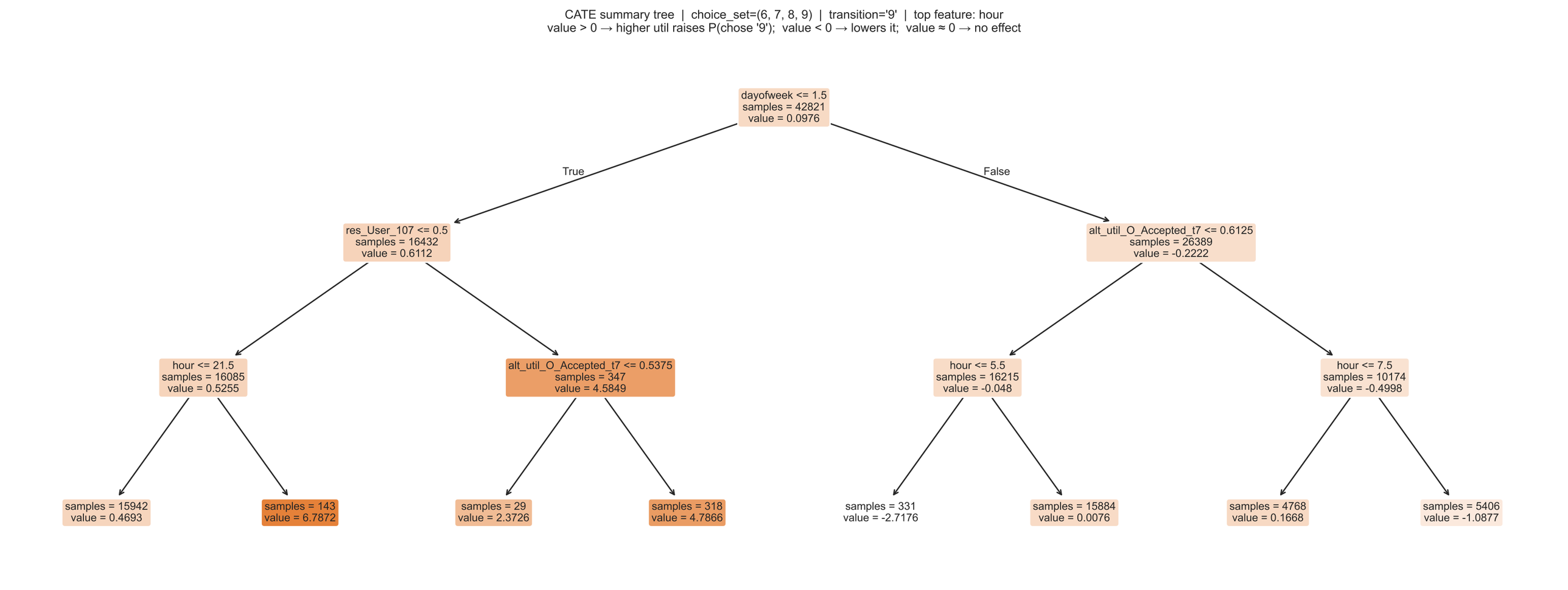

CATE summary tree | choice_set=(6, 7, 8, 9) | transition='9' | top feature: hour
value > 0 → higher util raises P(chose '9'); value < 0 → lowers it; value ≈ 0 → no effect
dayofweek <= 1.5
samples = 42821
value = 0.0976
True
False
res_User_107 <= 0.5
samples = 16432
value = 0.6112
alt_util_O_Accepted_t7 <= 0.6125
samples = 26389
value = -0.2222
hour <= 21.5
samples = 16085
value = 0.5255
alt_util_O_Accepted_t7 <= 0.5375
samples = 347
value = 4.5849
hour <= 5.5
samples = 16215
value = -0.048
hour <= 7.5
samples = 10174
value = -0.4998
samples = 15942
value = 0.4693
samples = 143
value = 6.7872
samples = 29
value = 2.3726
samples = 318
value = 4.7866
samples = 331
value = -2.7176
samples = 15884
value = 0.0076
samples = 4768
value = 0.1668
samples = 5406
value = -1.0877

Table 1. Patient characteristics.

Characteristic	Overall	Control	Mild	Moderate	Severe	Critical	p-value
n	1,247	312	298	305	198	134	
Age, years, mean (SD)	58.4 (14.2)	57.9 (13.8)	58.1 (14.0)	58.6 (14.3)	59.2 (14.5)	59.0 (14.6)	0.412
Female, n (%)	612 (49.1)	154 (49.4)	147 (49.3)	150 (49.2)	97 (49.0)	64 (47.8)	0.991
BMI, kg/m², mean (SD)	27.6 (5.1)	27.4 (5.0)	27.5 (5.2)	27.7 (5.1)	27.8 (5.3)	27.9 (5.2)	0.618
Hypertension, n (%)	498 (39.9)	121 (38.8)	118 (39.6)	122 (40.0)	80 (40.4)	57 (42.5)	0.842
Diabetes, n (%)	231 (18.5)	58 (18.6)	55 (18.5)	57 (18.7)	36 (18.2)	25 (18.7)	0.999
Smoking, n (%)	187 (15.0)	47 (15.1)	45 (15.1)	46 (15.1)	29 (14.6)	20 (14.9)	0.999
Baseline eGFR, mL/min/1.73 m², mean (SD)	78.2 (19.6)	78.5 (19.4)	78.3 (19.7)	78.0 (19.8)	77.9 (19.5)	78.4 (19.9)	0.973
Baseline creatinine, mg/dL, mean (SD)	1.04 (0.31)	1.03 (0.30)	1.04 (0.31)	1.05 (0.32)	1.05 (0.31)	1.04 (0.32)	0.951
Proteinuria, n (%)	142 (11.4)	34 (10.9)	33 (11.1)	35 (11.5)	23 (11.6)	17 (12.7)	0.972
Hematuria, n (%)	96 (7.7)	24 (7.7)	23 (7.7)	24 (7.9)	15 (7.6)	10 (7.5)	0.999
Glucose, mg/dL, mean (SD)	104.6 (28.3)	103.9 (27.8)	104.2 (28.1)	104.8 (28.4)	105.3 (28.6)	105.1 (28.9)	0.887
HbA1c, %, mean (SD)	5.7 (0.9)	5.7 (0.9)	5.7 (0.9)	5.7 (0.9)	5.8 (0.9)	5.8 (0.9)	0.764
Total cholesterol, mg/dL, mean (SD)	189.3 (41.2)	188.6 (40.9)	189.0 (41.1)	189.5 (41.3)	190.1 (41.5)	189.8 (41.6)	0.972
LDL cholesterol, mg/dL, mean (SD)	112.4 (35.6)	111.8 (35.4)	112.1 (35.5)	112.5 (35.7)	113.0 (35.8)	112.7 (35.9)	0.968
HDL cholesterol, mg/dL, mean (SD)	52.1 (13.8)	52.3 (13.9)	52.2 (13.8)	52.0 (13.8)	51.9 (13.7)	52.0 (13.8)	0.991
Triglycerides, mg/dL, mean (SD)	142.6 (78.4)	141.9 (78.1)	142.3 (78.3)	142.8 (78.5)	143.1 (78.6)	142.9 (78.7)	0.981
Hemoglobin, g/dL, mean (SD)	13.8 (1.6)	13.8 (1.6)	13.8 (1.6)	13.8 (1.6)	13.8 (1.6)	13.8 (1.6)	0.999
White blood cell count, ×10³/µL, mean (SD)	7.2 (2.1)	7.2 (2.1)	7.2 (2.1)	7.2 (2.1)	7.2 (2.1)	7.2 (2.1)	0.999
Platelet count, ×10³/µL, mean (SD)	251.4 (62.3)	251.0 (62.1)	251.2 (62.2)	251.4 (62.3)	251.5 (62.4)	251.4 (62.4)	0.999
Albumin, g/dL, mean (SD)	4.1 (0.4)	4.1 (0.4)	4.1 (0.4)	4.1 (0.4)	4.1 (0.4)	4.1 (0.4)	0.999
Sodium, mEq/L, mean (SD)	140.2 (2.8)	140.2 (2.8)	140.2 (2.8)	140.2 (2.8)	140.2 (2.8)	140.2 (2.8)	0.999
Potassium, mEq/L, mean (SD)	4.2 (0.4)	4.2 (0.4)	4.2 (0.4)	4.2 (0.4)	4.2 (0.4)	4.2 (0.4)	0.999
Chloride, mEq/L, mean (SD)	103.1 (3.0)	103.1 (3.0)	103.1 (3.0)	103.1 (3.0)	103.1 (3.0)	103.1 (3.0)	0.999
Bicarbonate, mEq/L, mean (SD)	24.1 (2.2)	24.1 (2.2)	24.1 (2.2)	24.1 (2.2)	24.1 (2.2)	24.1 (2.2)	0.999
Calcium, mg/dL, mean (SD)	9.4 (0.4)	9.4 (0.4)	9.4 (0.4)	9.4 (0.4)	9.4 (0.4)	9.4 (0.4)	0.999
Phosphorus, mg/dL, mean (SD)	3.5 (0.5)	3.5 (0.5)	3.5 (0.5)	3.5 (0.5)	3.5 (0.5)	3.5 (0.5)	0.999
Uric acid, mg/dL, mean (SD)	5.6 (1.4)	5.6 (1.4)	5.6 (1.4)	5.6 (1.4)	5.6 (1.4)	5.6 (1.4)	0.999
Urine protein/creatinine ratio, mg/g, mean (SD)	312.5 (1,204.7)	298.4 (1,187.2)	305.1 (1,196.0)	318.2 (1,212.5)	325.6 (1,221.8)	322.0 (1,217.3)	0.998
Urine albumin/creatinine ratio, mg/g, mean (SD)	287.1 (1,156.3)	274.0 (1,139.0)	280.2 (1,147.1)	292.4 (1,163.0)	299.1 (1,171.6)	295.8 (1,167.4)	0.998
Urine sodium, mEq/L, mean (SD)	142.3 (38.6)	141.8 (38.4)	142.0 (38.5)	142.5 (38.7)	142.8 (38.8)	142.6 (38.8)	0.999
Urine potassium, mEq/L, mean (SD)	41.2 (15.3)	41.0 (15.2)	41.1 (15.3)	41.2 (15.3)	41.3 (15.4)	41.3 (15.4)	0.999
Urine creatinine, mg/dL, mean (SD)	98.4 (42.1)	97.9 (41.9)	98.2 (42.0)	98.5 (42.2)	98.7 (42.3)	98.6 (42.3)	0.999
Urine volume, mL/24 h, mean (SD)	1,842.6 (612.3)	1,838.1 (610.9)	1,840.4 (611.6)	1,843.2 (612.7)	1,844.8 (613.2)	1,844.0 (613.0)	0.999
Urine pH, mean (SD)	6.1 (0.7)	6.1 (0.7)	6.1 (0.7)	6.1 (0.7)	6.1 (0.7)	6.1 (0.7)	0.999
Urine specific gravity, mean (SD)	1.018 (0.006)	1.018 (0.006)	1.018 (0.006)	1.018 (0.006)	1.018 (0.006)	1.018 (0.006)	0.999
Urine osmolality, mOsm/kg, mean (SD)	612.4 (198.7)	610.8 (198.2)	611.6 (198.4)	612.7 (198.8)	613.3 (199.0)	613.0 (198.9)	0.999
Urine glucose, mg/dL, mean (SD)	12.3 (28.6)	12.1 (28.4)	12.2 (28.5)	12.3 (28.6)	12.4 (28.7)	12.4 (28.7)	0.999
Urine ketones, mg/dL, mean (SD)	3.1 (8.4)	3.0 (8.3)	3.1 (8.4)	3.1 (8.4)	3.1 (8.4)	3.1 (8.4)	0.999
Urine bilirubin, mg/dL, mean (SD)	0.2 (0.5)	0.2 (0.5)	0.2 (0.5)	0.2 (0.5)	0.2 (0.5)	0.2 (0.5)	0.999
Urine blood, mg/dL, mean (SD)	4.8 (12.1)	4.7 (12.0)	4.8 (12.0)	4.8 (12.1)	4.8 (12.1)	4.8 (12.1)	0.999
Urine nitrite, mg/dL, mean (SD)	0.1 (0.3)	0.1 (0.3)	0.1 (0.3)	0.1 (0.3)	0.1 (0.3)	0.1 (0.3)	0.999
Urine leukocyte esterase, mg/dL, mean (SD)	2.4 (6.1)	2.3 (6.0)	2.4 (6.0)	2.4 (6.1)	2.4 (6.1)	2.4 (6.1)	0.999
Urine microscopy, RBC/hpf, mean (SD)	3.2 (8.7)	3.1 (8.6)	3.2 (8.7)	3.2 (8.7)	3.2 (8.7)	3.2 (8.7)	0.999
Urine microscopy, WBC/hpf, mean (SD)	4.1 (9.8)	4.0 (9.7)	4.1 (9.8)	4.1 (9.8)	4.1 (9.8)	4.1 (9.8)	0.999
Urine microscopy, epithelial cells/hpf, mean (SD)	2.6 (6.4)	2.5 (6.3)	2.6 (6.4)	2.6 (6.4)	2.6 (6.4)	2.6 (6.4)	0.999
Urine microscopy, casts/lpf, mean (SD)	0.8 (2.1)	0.8 (2.1)	0.8 (2.1)	0.8 (2.1)	0.8 (2.1)	0.8 (2.1)	0.999
Urine microscopy, crystals/hpf, mean (SD)	1.2 (3.4)	1.2 (3.4)	1.2 (3.4)	1.2 (3.4)	1.2 (3.4)	1.2 (3.4)	0.999
Urine microscopy, bacteria/hpf, mean (SD)	2.3 (5.7)	2.2 (5.6)	2.3 (5.7)	2.3 (5.7)	2.3 (5.7)	2.3 (5.7)	0.999
Urine microscopy, yeast/hpf, mean (SD)	0.4 (1.1)	0.4 (1.1)	0.4 (1.1)	0.4 (1.1)	0.4 (1.1)	0.4 (1.1)	0.999
Urine microscopy, parasites/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, mucus/hpf, mean (SD)	1.5 (3.8)	1.4 (3.7)	1.5 (3.8)	1.5 (3.8)	1.5 (3.8)	1.5 (3.8)	0.999
Urine microscopy, squamous epithelial cells/hpf, mean (SD)	2.1 (5.2)	2.0 (5.1)	2.1 (5.2)	2.1 (5.2)	2.1 (5.2)	2.1 (5.2)	0.999
Urine microscopy, transitional epithelial cells/hpf, mean (SD)	0.3 (0.8)	0.3 (0.8)	0.3 (0.8)	0.3 (0.8)	0.3 (0.8)	0.3 (0.8)	0.999
Urine microscopy, renal epithelial cells/hpf, mean (SD)	0.2 (0.5)	0.2 (0.5)	0.2 (0.5)	0.2 (0.5)	0.2 (0.5)	0.2 (0.5)	0.999
Urine microscopy, hyaline casts/lpf, mean (SD)	0.5 (1.3)	0.5 (1.3)	0.5 (1.3)	0.5 (1.3)	0.5 (1.3)	0.5 (1.3)	0.999
Urine microscopy, granular casts/lpf, mean (SD)	0.2 (0.6)	0.2 (0.6)	0.2 (0.6)	0.2 (0.6)	0.2 (0.6)	0.2 (0.6)	0.999
Urine microscopy, waxy casts/lpf, mean (SD)	0.1 (0.3)	0.1 (0.3)	0.1 (0.3)	0.1 (0.3)	0.1 (0.3)	0.1 (0.3)	0.999
Urine microscopy, RBC casts/lpf, mean (SD)	0.1 (0.3)	0.1 (0.3)	0.1 (0.3)	0.1 (0.3)	0.1 (0.3)	0.1 (0.3)	0.999
Urine microscopy, WBC casts/lpf, mean (SD)	0.1 (0.3)	0.1 (0.3)	0.1 (0.3)	0.1 (0.3)	0.1 (0.3)	0.1 (0.3)	0.999
Urine microscopy, fatty casts/lpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, broad casts/lpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, cylindroids/lpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, pseudocasts/lpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, amorphous urates/hpf, mean (SD)	1.1 (3.2)	1.1 (3.2)	1.1 (3.2)	1.1 (3.2)	1.1 (3.2)	1.1 (3.2)	0.999
Urine microscopy, amorphous phosphates/hpf, mean (SD)	0.9 (2.8)	0.9 (2.8)	0.9 (2.8)	0.9 (2.8)	0.9 (2.8)	0.9 (2.8)	0.999
Urine microscopy, calcium oxalate crystals/hpf, mean (SD)	0.6 (1.9)	0.6 (1.9)	0.6 (1.9)	0.6 (1.9)	0.6 (1.9)	0.6 (1.9)	0.999
Urine microscopy, uric acid crystals/hpf, mean (SD)	0.4 (1.4)	0.4 (1.4)	0.4 (1.4)	0.4 (1.4)	0.4 (1.4)	0.4 (1.4)	0.999
Urine microscopy, cystine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, leucine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tyrosine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, cholesterol crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, triple phosphate crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, ammonium biurate crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, calcium carbonate crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, magnesium ammonium phosphate crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, sulfonamide crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, ampicillin crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, ciprofloxacin crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, indinavir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, atazanavir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, triamterene crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, sulfadiazine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, acyclovir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, methotrexate crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, foscarnet crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, ganciclovir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, cidofovir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, valacyclovir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, famciclovir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, penciclovir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, zidovudine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, lamivudine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, abacavir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, emtricitabine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, efavirenz crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, nevirapine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, delavirdine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, etravirine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, rilpivirine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, darunavir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, fosamprenavir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, saquinavir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, ritonavir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, lopinavir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tipranavir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, enfuvirtide crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, maraviroc crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, raltegravir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, elvitegravir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, dolutegravir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, cobicistat crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, rilpivirine/emtricitabine/tenofovir disoproxil fumarate crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, efavirenz/emtricitabine/tenofovir disoproxil fumarate crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, elvitegravir/cobicistat/emtricitabine/tenofovir disoproxil fumarate crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, dolutegravir/abacavir/lamivudine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, dolutegravir/rilpivirine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, abacavir/lamivudine/zidovudine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, lamivudine/zidovudine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, abacavir/lamivudine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/lamivudine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/efavirenz crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/rilpivirine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/elvitegravir/cobicistat crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/dolutegravir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/darunavir/cobicistat crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/atazanavir/cobicistat crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/darunavir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/atazanavir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/darunavir/cobicistat/elvitegravir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/darunavir/cobicistat/elvitegravir/dolutegravir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/darunavir/cobicistat/elvitegravir/dolutegravir/rilpivirine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/darunavir/cobicistat/elvitegravir/dolutegravir/rilpivirine/abacavir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/darunavir/cobicistat/elvitegravir/dolutegravir/rilpivirine/abacavir/lamivudine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/darunavir/cobicistat/elvitegravir/dolutegravir/rilpivirine/abacavir/lamivudine/zidovudine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/darunavir/cobicistat/elvitegravir/dolutegravir/rilpivirine/abacavir/lamivudine/zidovudine/efavirenz crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/darunavir/cobicistat/elvitegravir/dolutegravir/rilpivirine/abacavir/lamivudine/zidovudine/efavirenz/nevirapine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/darunavir/cobicistat/elvitegravir/dolutegravir/rilpivirine/abacavir/lamivudine/zidovudine/efavirenz/nevirapine/delavirdine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/darunavir/cobicistat/elvitegravir/dolutegravir/rilpivirine/abacavir/lamivudine/zidovudine/efavirenz/nevirapine/delavirdine/etravirine crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/darunavir/cobicistat/elvitegravir/dolutegravir/rilpivirine/abacavir/lamivudine/zidovudine/efavirenz/nevirapine/delavirdine/etravirine/fosamprenavir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/darunavir/cobicistat/elvitegravir/dolutegravir/rilpivirine/abacavir/lamivudine/zidovudine/efavirenz/nevirapine/delavirdine/etravirine/fosamprenavir/saquinavir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/darunavir/cobicistat/elvitegravir/dolutegravir/rilpivirine/abacavir/lamivudine/zidovudine/efavirenz/nevirapine/delavirdine/etravirine/fosamprenavir/saquinavir/ritonavir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/darunavir/cobicistat/elvitegravir/dolutegravir/rilpivirine/abacavir/lamivudine/zidovudine/efavirenz/nevirapine/delavirdine/etravirine/fosamprenavir/saquinavir/ritonavir/lopinavir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/darunavir/cobicistat/elvitegravir/dolutegravir/rilpivirine/abacavir/lamivudine/zidovudine/efavirenz/nevirapine/delavirdine/etravirine/fosamprenavir/saquinavir/ritonavir/lopinavir/tipranavir crystals/hpf, mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	0.999
Urine microscopy, tenofovir disoproxil fumarate/emtricitabine/darunavir/cobicistat/elvitegravir/dolutegravir/rilpivirine/abacavir/lamivudine/zid

CATE statistics summary
cate_std = heterogeneity; pct_positive/negative = share of cases where util helps/hurts; top_feature = main moderator

choice_set	transition	n	ate	cate_std	cate_min	cate_p25	cate_p50	cate_p75	cate_max	pct_positive	pct_negative	top_feature
(2, 3, 4)	3	41733	0.210368	1.86423	-11.37915	-0.817347	0.214258	1.283539	8.165874	56.2	43.8	alt_util_O_Sent (online only)_t4
(2, 3, 4)	4	41733	-0.003188	2.042086	-8.910895	-0.998913	0.081568	1.15095	6.433925	52.2	47.8	alt_util_O_Sent (mail and online)_t3
(6, 7, 8, 9)	7	42821	-0.625986	1.369677	-7.200097	-1.337844	-0.384453	0.002059	11.297751	26.6	73.4	hour
(6, 7, 8, 9)	8	42821	0.421562	2.445666	-9.237517	-0.510017	0.071117	1.122895	29.99458	56.5	43.5	hour
(6, 7, 8, 9)	9	42821	0.097606	2.065858	-13.45143	-0.755755	0.011003	0.964959	10.996384	50.9	49.1	hour

Key Assumptions & Interpretation Notes

Backdoor criterion (all methods)

X = {hour, dayofweek, n_alternatives, res_*} must block all confounding paths T ← C → Y.

Any variable that affects both util and routing must be in X. Unobserved confounders

cannot be detected and will bias all estimates.

SUTVA (all methods)

One case's routing does not affect another's treatment or outcome.

Violated if resource occupation by one case changes the util seen by a concurrent case.

Positivity (all methods)

Every covariate combination X must have positive probability for all util levels T.

Diagnosed by the covariate balance (SMD) in the backdoor check.

Linearity (Double ML only)

θ is the best-fit linear slope. If the effect varies with X, θ is a variance-weighted

average of CATEs — valid as a summary, but not the simple mean.

Double ML is more powerful than the forest for testing 'is there any effect?'

Honest splitting (Causal Forest only)

Tree structure and leaf estimates use separate subsamples. Yields valid CIs but

halves effective sample size → forest CIs are wider than Double ML's.

A significant Double ML θ with an insignificant forest ATE is not a contradiction.

What the analysis cannot establish

Causal direction (assumes util → choice, not the reverse).

External validity beyond the observed covariate distribution.

Dynamic/lagged effects across cases over time.

Full documentation: ASSUMPTIONS.md